Q NO 1 :-

$$2x + z - y = 4$$
$$4x - 6y + 2z = 3$$

Sol:-

$$n_1 = \langle 2, -1, 1 \rangle$$
$$n_2 = \langle 4, -6, 2 \rangle$$

$$n_1 \times n_2 = \begin{vmatrix} i & j & k \\ 2 & -1 & 1 \\ 4 & -6 & 2 \end{vmatrix}$$

$$n_1 \times n_2 = i(-2+6) - j(4-4) + k(-12+4)$$

$$= 4i - j(0) + k(-8)$$

$$= \langle 4, 0, -8 \rangle$$

Now:-

$$2x - y + z = 4 \quad (1)$$
$$4x - 6y + 2z = 3 \quad (2)$$

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By taking 2-intercept :-

$$2x - y = 4 \quad (1)$$

$$4x - 6y = 3 \quad (2)$$

multiply eq (1) by 2 :-

$$\cancel{4x} - 2y = 8$$

$$-\cancel{4x} \mp 6y = -3$$

$$4y = 5$$

$$y = 5/4$$

By putting y in equation:-

$$4x - \cancel{6}^{3}\left(\frac{5}{\cancel{4}_{2}}\right) = 3$$

$$4x - \frac{15}{2} = 3$$

$$\frac{2(4x) - 15}{2} = 3$$

Ans

$$2(4x) - 15 = 6$$

$$2(4x) = 6 + 15$$

$$4x = \frac{(6+15)}{2}$$

$$x = \left(\frac{6+15}{2}\right)$$

$$= \left(\frac{21}{2}\right) \times 4$$

$$\boxed{x = \frac{21}{8}}$$

So $\langle \frac{21}{8}, \frac{5}{4}, 0 \rangle$

Parameterization:-

$$\left.\begin{aligned} x &= 21/8 + 4t \\ y &= 5/4 + 0t \\ z &= 0 - 8t \end{aligned}\right\} \text{From} \left\{\begin{aligned} x &= x_0 + v_1 t \\ y &= y_0 + v_2 t \\ z &= z_0 + v_3 t \end{aligned}\right\}$$

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(b)

$\vec{a}(t) = -5\sin t - 5\cos t\, j + 2K$

$V(0) = 5\,\vec{i} + \vec{j}$

Sol:-

$v(t) = \int \vec{a}(t)\, dt$

$\vec{v}(t) = \int (-5\sin t\, i - 5\cos t\, j + 2k)\, dt$

$= -5\int \sin t\, i\, dt - 5\int \cos t\, j + 2\int K\, dt$

$= -5(-\cos t\, i) - 5\sin t\, j + 2tK + C$

$= 5\cos t\, i - 5\sin t\, j + 2tK + C$

$V(0) = 5\,i + j$

$\vec{v}(0) = 5\cos(0)\,\vec{i} - 5\sin(0)\, j + 2(0) + C$

$\cancel{5i} + j = \cancel{5i} + C$

$\boxed{C = j}$

$= 5\cos t\, i - 5\sin t\, j + 2tK + j$

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